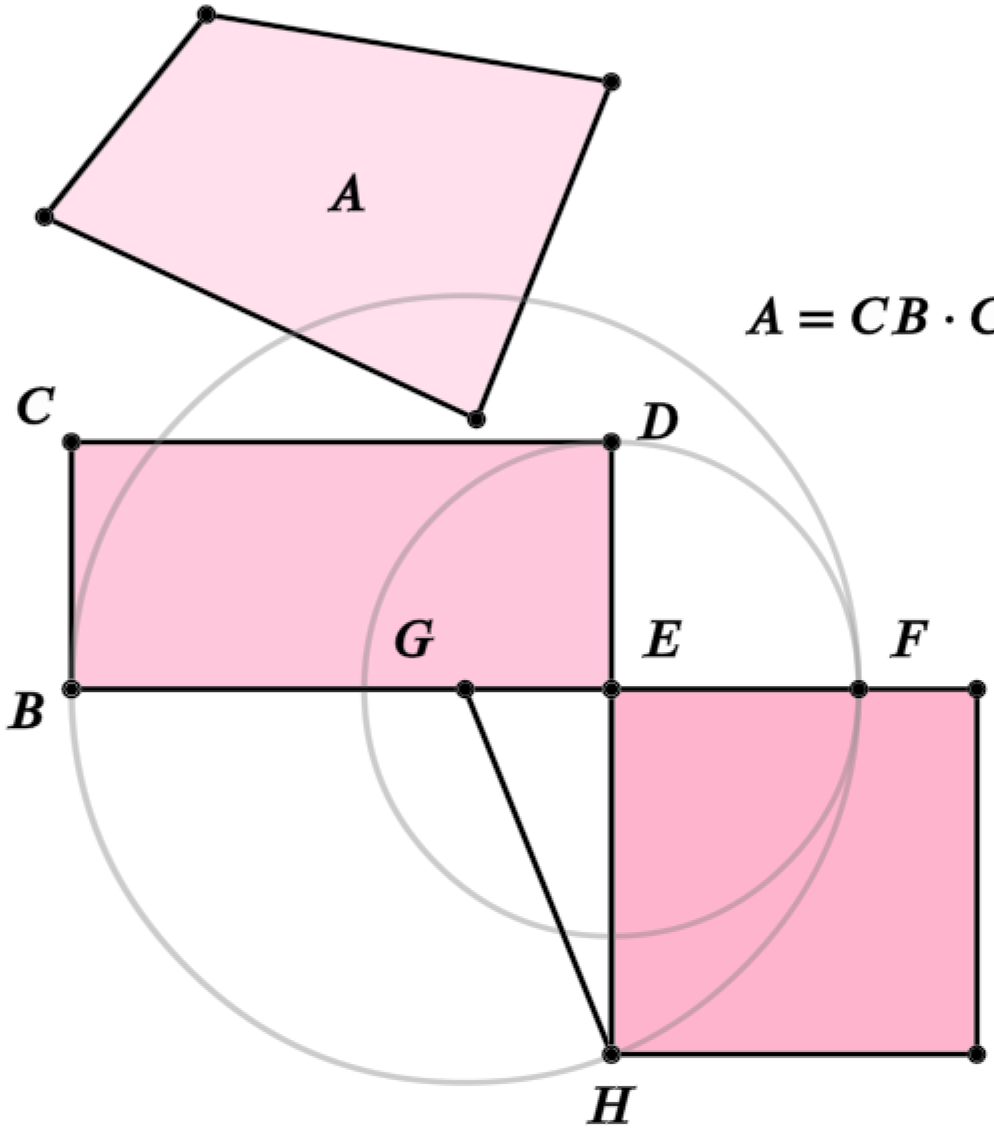
Euclid's Elements

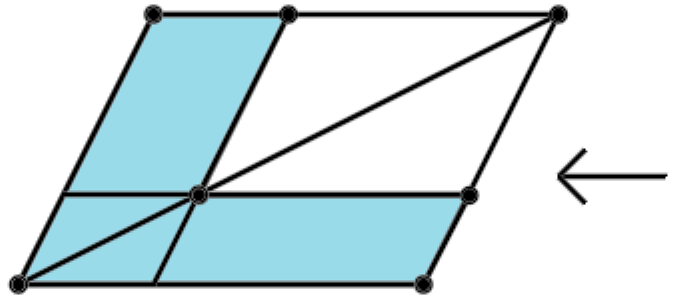
Book II



$$A = CB \cdot CD = (EH)^2$$

It is a remarkable fact in the history of geometry, that the Elements of Euclid, written two thousand years ago, are still regarded by many as the best introduction to the mathematical sciences.

- Florian Cajori,
A History of Mathematics (1893)



Definitions:

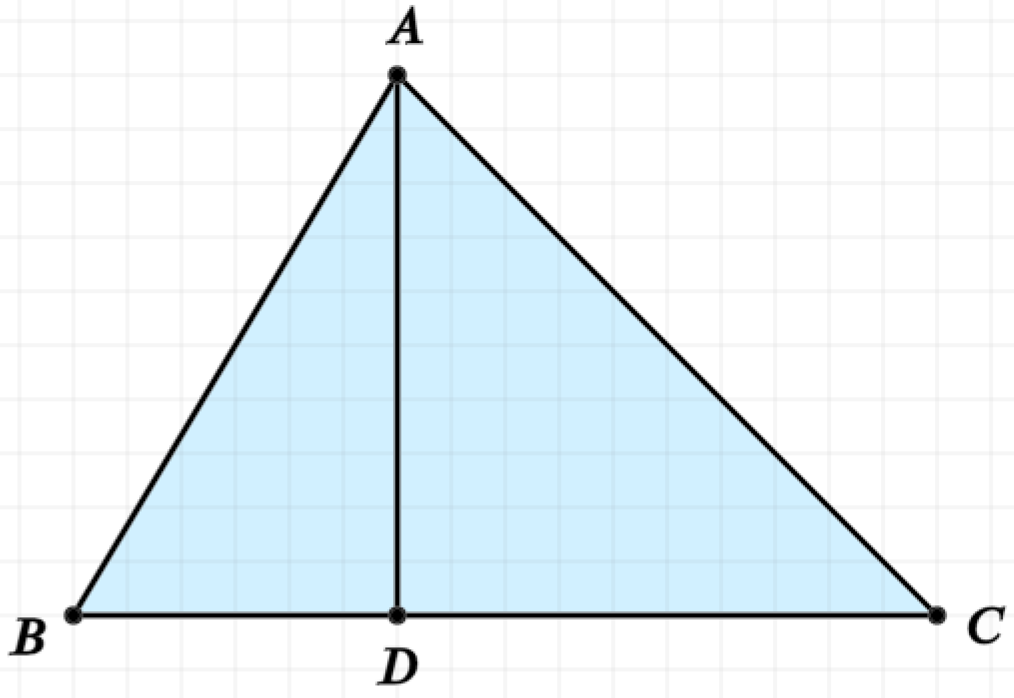
Any rectangular parallelogram is said to be contained by the two straight lines containing the right angle.

And in any parallelogrammic area let any one whatever of the parallelograms about its diameter with the two complements be called a gnomon.

Table of Contents - Book 2

Proposition 13 of Book 2

In acute-angled triangles the square on the side subtending the acute angle is less than the squares on the sides containing the acute angle by twice the rectangle contained by one of the sides about the acute angle, namely that on which the perpendicular falls, and the straight line cut off within by the perpendicular towards the acute angle.



In other words

Given an acute triangle ABC where the acute angle is at point B and the base is BC:

The square of AC equals the sum of the squares of AB and BC, less...

... twice the rectangle formed by base BC and the portion of the base, BD, where BD is the length from the base to the perpendicular from A to the base

$$AC^2 = AB^2 + BC^2 - 2 \cdot BD \cdot BC$$

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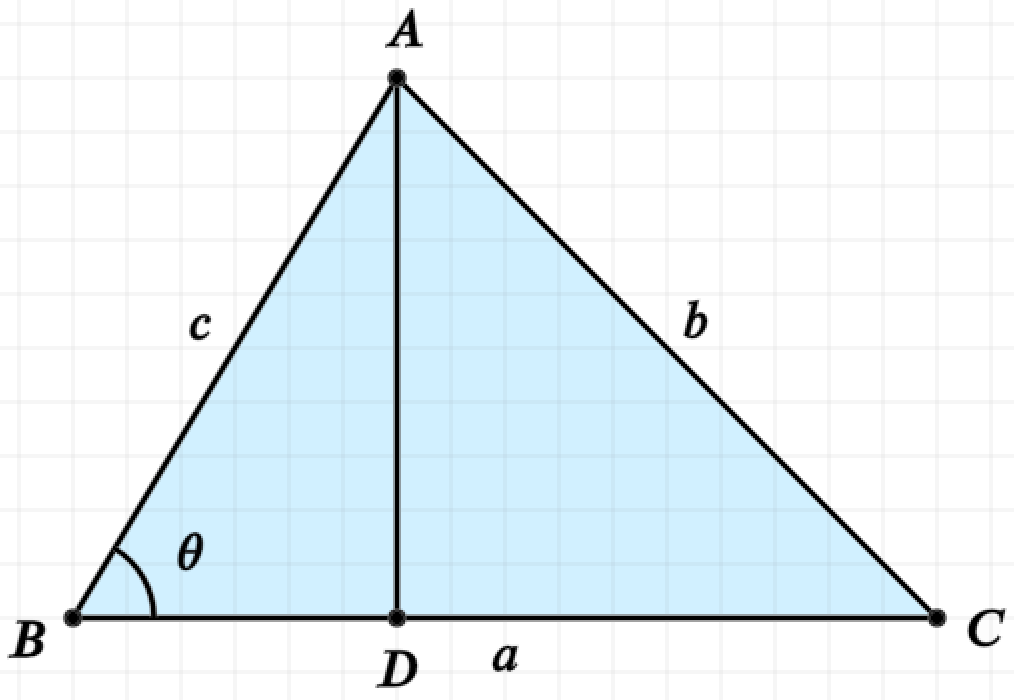

In other words

Given an acute triangle ABC where the acute angle is at point B and the base is BC:

The square of AC equals the sum of the squares of AB and BC, less...

... twice the rectangle formed by base BC and the portion of the base, BD, where BD is the length from the base to the perpendicular from A to the base

$$AC^2 = AB^2 + BC^2 - 2 \cdot BD \cdot BC$$

Or... the cosine law

$$BC = a, \quad AB = c, \quad AC = b$$

$$BD = c \cos \alpha$$

$$b^2 = c^2 + a^2 - 2ac \cos \theta$$

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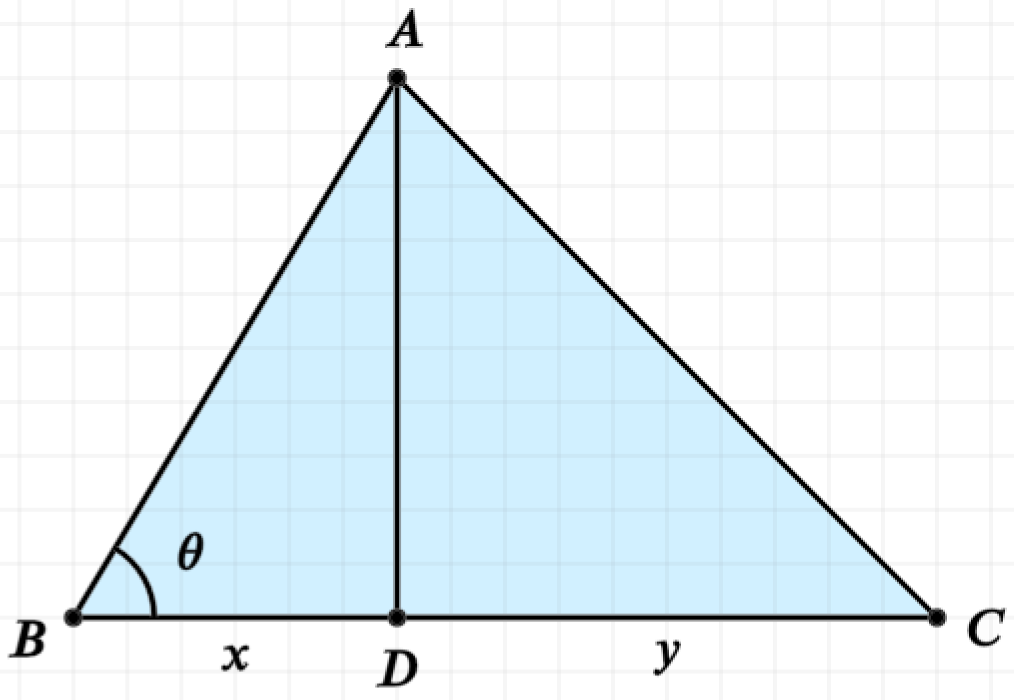

$$BC^2 + BD^2 = DC^2 + 2 \cdot BD \cdot BC$$

$$(x+y)^2 + x^2 = y^2 + 2x(x+y)$$

In other words

Given an acute triangle ABC where the acute angle is at point B and the base is BC:

$$AC^2 = AB^2 + BC^2 - 2 \cdot BD \cdot BC$$

Proof

The line BC is cut at a point D, and the thus the sum of the squares of BC and BD is equal to the square of DC plus twice the rectangle formed by BC and BD (II.7)

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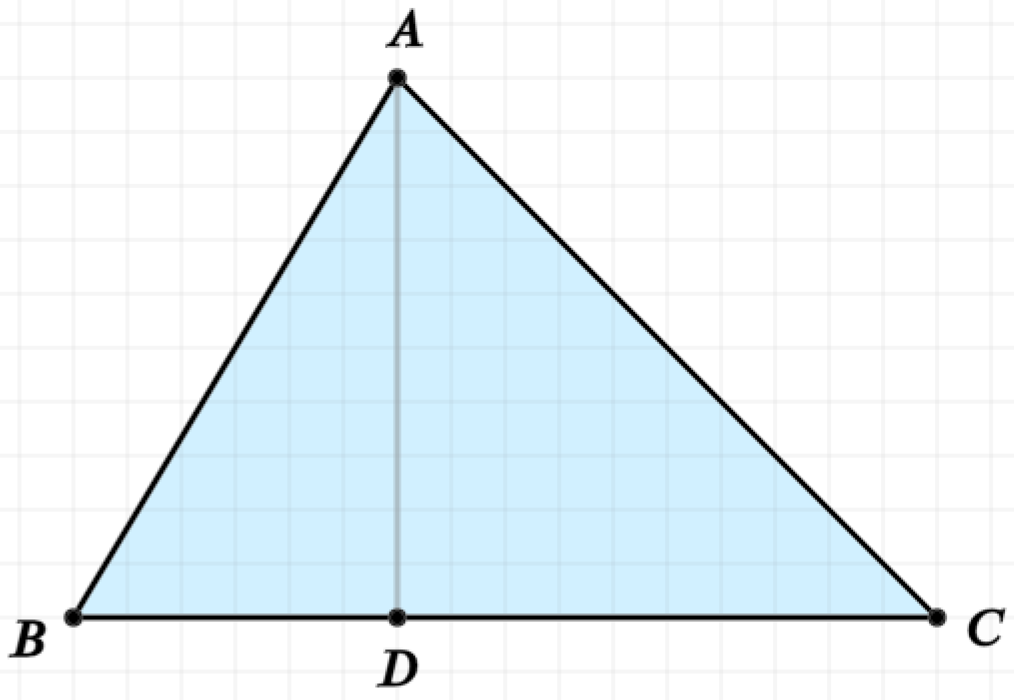

$$BC^2 + BD^2 = DC^2 + 2 \cdot BD \cdot BC$$

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Add the square of AD to both sides of the equality

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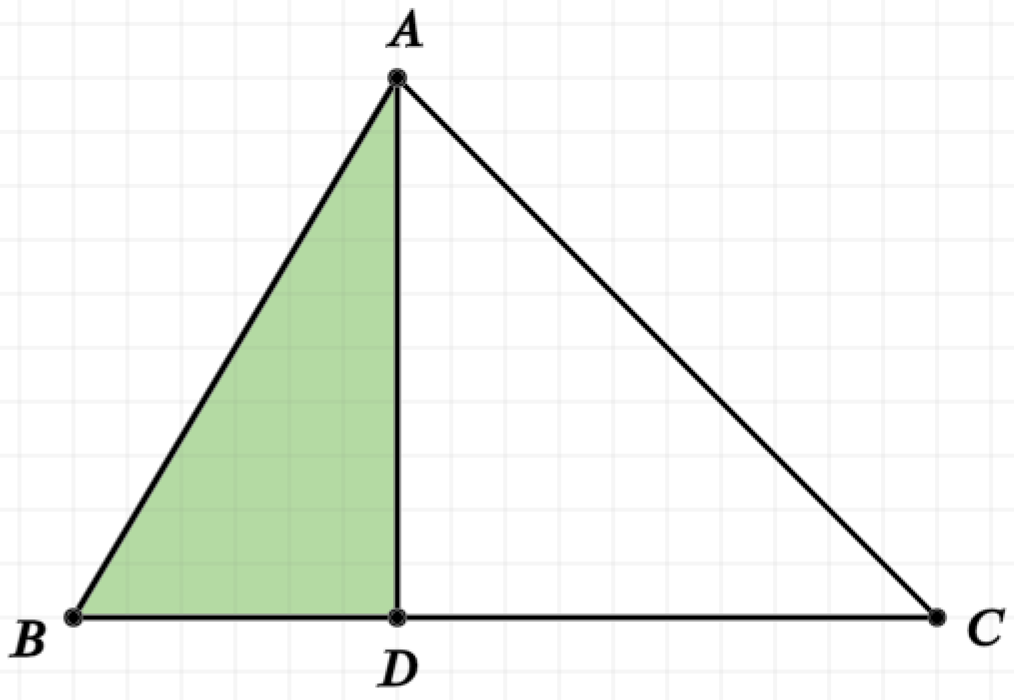

$$BC^2 + BD^2 = DC^2 + 2 \cdot BD \cdot BC$$

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The squares of BD and AD equals the square of AB (I.47)

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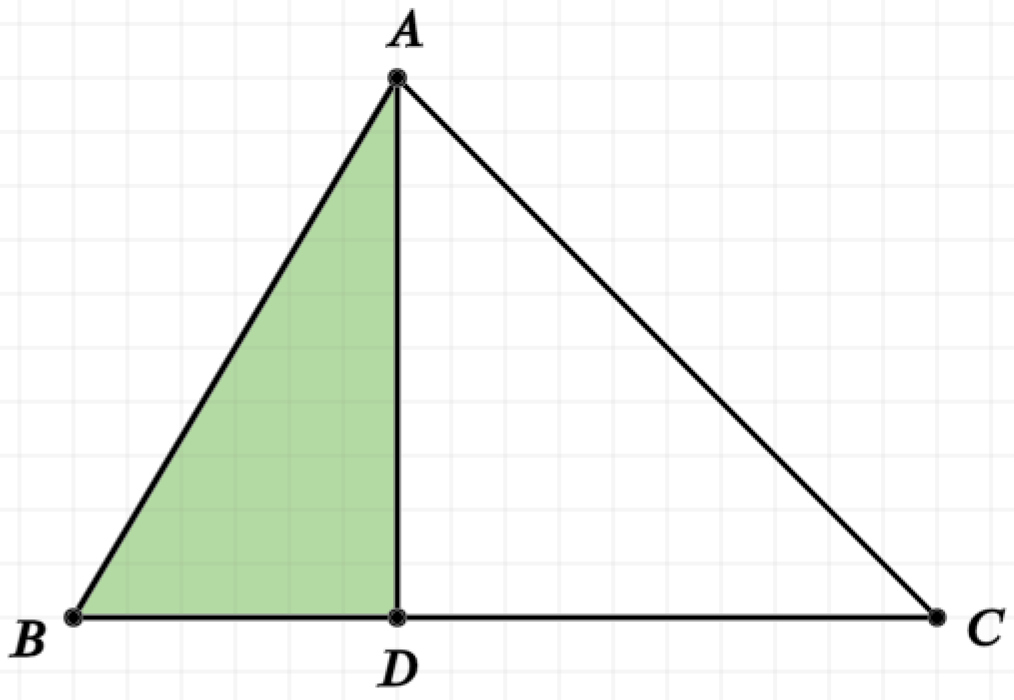

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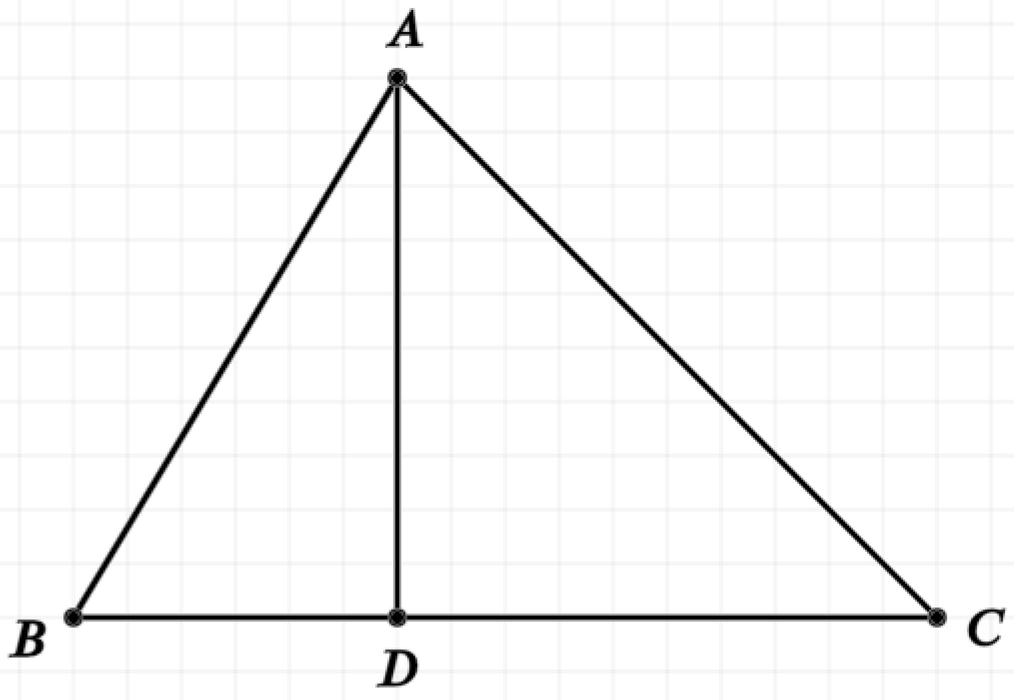

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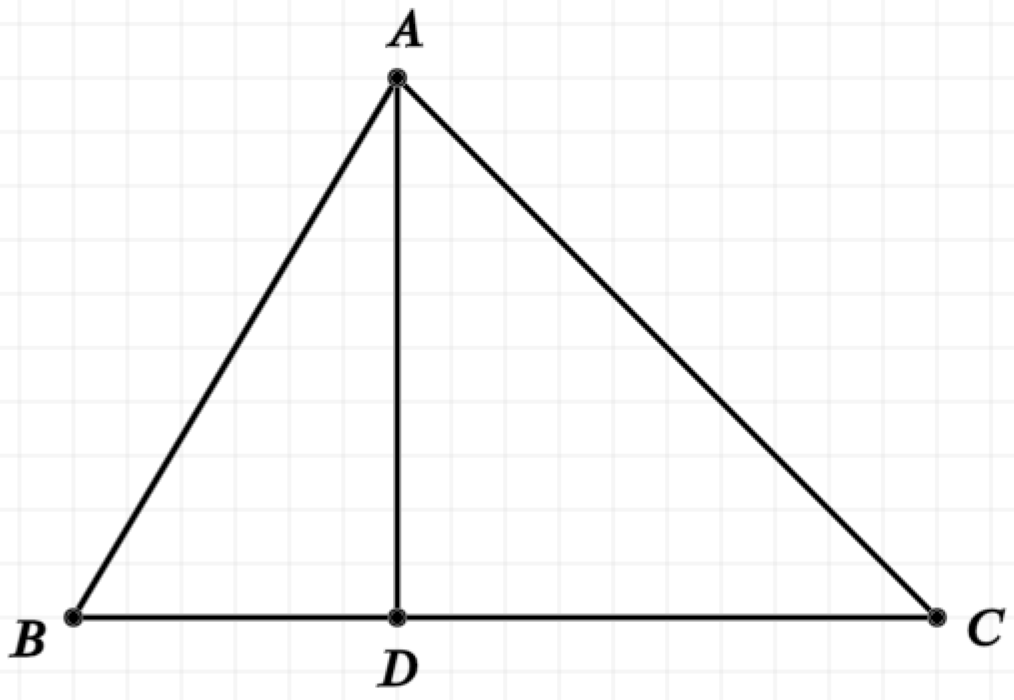

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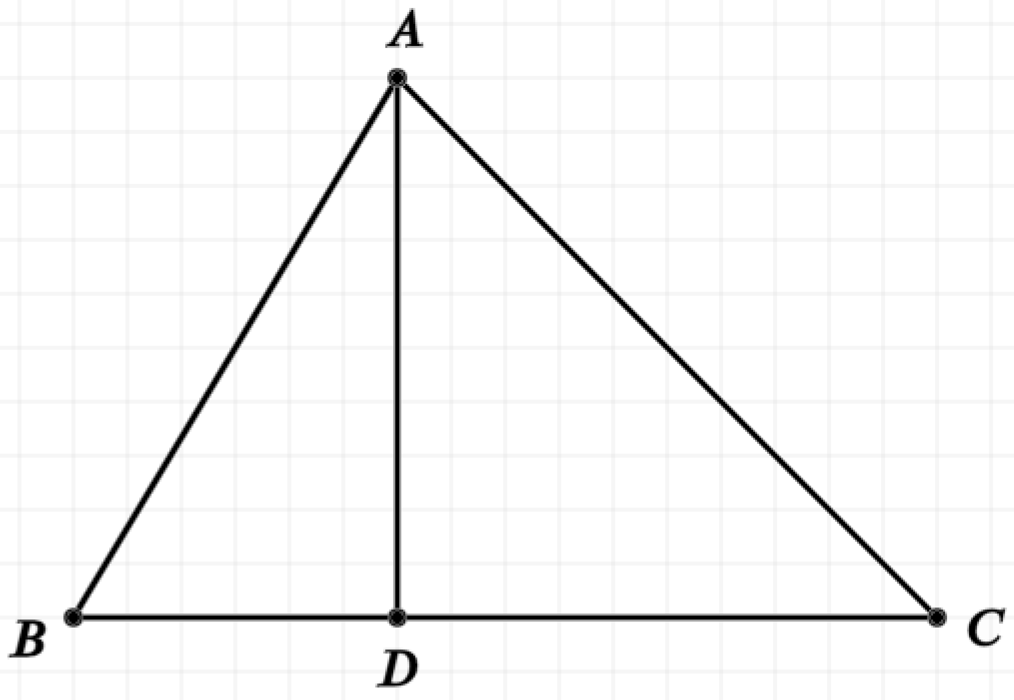

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The squares of AD and DC equals the square of AC (I.47)

Thus the square of AC is equal to the sum of the squares of BC and AB, less the rectangle formed by BC,BD

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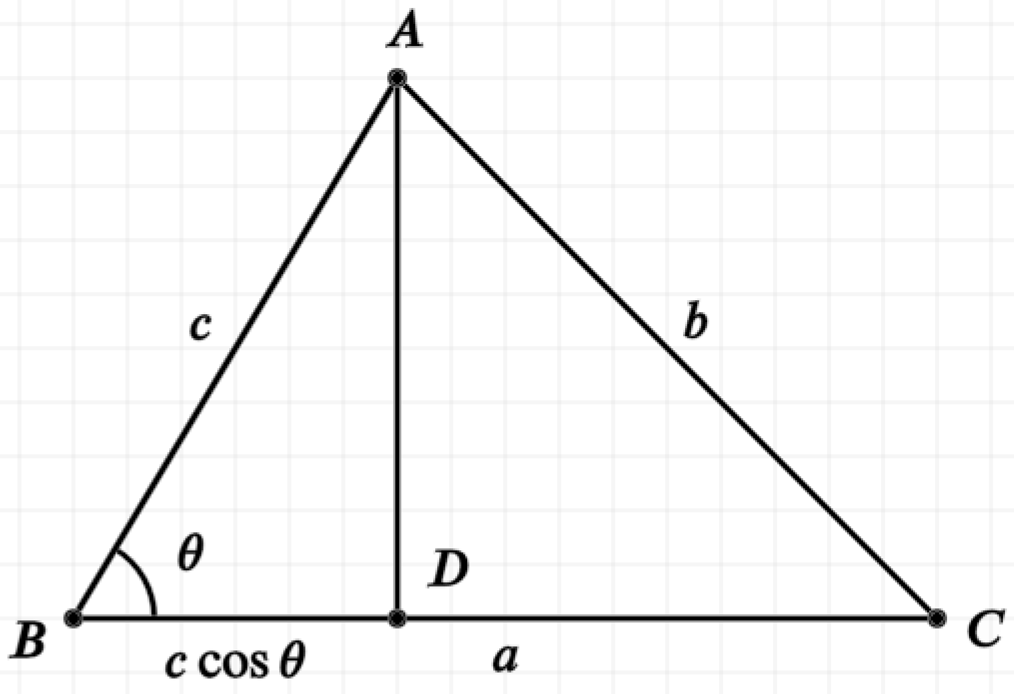

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